Chapter 2

Review of Statistics

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MGMTMSA 400

Population and Sample

- The **population** is the full group we want to understand. A **sample** is a selected subset of that population
- **Data** are the measurements we collect. Except for a few exceptions—such as census—we only observe data for samples, since collecting data from the whole population can be costly or difficult
- **Example:** Suppose Disney wants to know the average weekly Netflix viewing time among U.S. households with a Netflix subscription. Disney surveys 1000 of these households



Population Parameters and Sample Statistics

- A **parameter** describes the population. A **statistic** is a summary calculated from the sample
- We use sample statistics to **estimate** unknown population parameters



- If our sample average is 12 hours, we can estimate the population average as 12 hours. The true population average (parameter) is still unknown
- An estimate may be close to the true parameter or far from it. Its accuracy depends on factors like the sample size and how the sample was selected. We will discuss this in greater detail in **Estimation of the Mean and Standard Deviation**

Estimation and Prediction

- **Estimation** concerns an unknown population feature. **Prediction** concerns an outcome we have not observed

Estimation

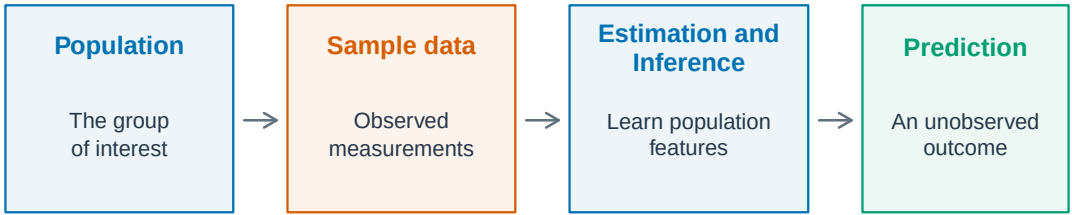
What is the average viewing time across the population?

Prediction

How many hours did ten unsampled household watch?

- Continuing the Netflix example, both answers might be **12 hours**: an estimate of the **population average**, or a prediction of **ten households' average viewing time**. Prediction is about an unobserved outcome for one or several individuals from the population, which may be in the future

How the Concepts Connect



A statistical model connects the data to our questions.

- **Chapter 2:** Use observations of an outcome, such as viewing hours, to estimate population features and make predictions
- **Later chapters:** Use additional information, such as household size and income, to improve estimation and prediction with regression

Road Map of Chapter 2

- Start with the **normal distribution** to introduce prediction and estimation
  - If the population mean and standard deviation are known, how do we make predictions? $\Longrightarrow$ **Normal Mean Problem**
  - If these parameters are unknown, how do we estimate them from a sample? $\Longrightarrow$ **Estimation of the Mean and Standard Deviation**
- The **Central Limit Theorem** helps us study sample means even when the original data are not normally distributed
- Use **confidence intervals** to describe uncertainty in estimates and **hypothesis tests** to evaluate claims about the population

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Normal Mean Problem

Normal Distribution

- Suppose a random variable Y follows a normal distribution:

$$Y \sim \mathcal{N}(\mu, \sigma^2)$$

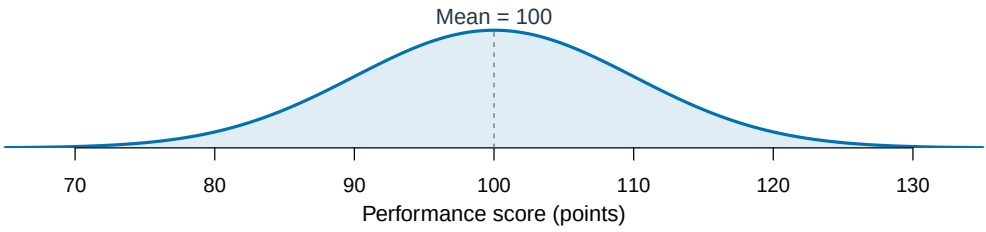
In this example, assume we know the values of μ and σ

- Fundamental properties of the normal distribution:
  - **Symmetric**: the left and right tails are mirror images, unlike those of a skewed distribution
  - **Unimodal**: the density has a single peak
  - **Thin tails**: $\text{Prob}[|Y - \mu| > 6\sigma] < 10^{-8}$
- The normal distribution can be used as a probability model for how data are generated

Example of Normal Distribution

- A **performance score** summarizes how quickly a chip completes a standard set of computing tasks; higher scores mean better performance
- Consider working chips of one design from a stable production process, tested under the same conditions. Small differences in manufacturing can lead to differences in performance
- Suppose their performance scores Y follow a normal distribution with a mean of **100 points** and a standard deviation of **10 points**:

$$Y \sim \mathcal{N}(100,\ 10^2), \qquad \mu = 100, \quad \sigma = 10$$



Samples from the Normal Distribution

- A sampling procedure is the procedure of selecting subsets (generating observations) from the population (underlying probability model)
- The standard sampling model is a "random" or independent and identically distributed sample (denoted **i.i.d**)

$$Y_1, Y_2, \ldots, Y_N \sim \text{i.i.d } \mathcal{N}(\mu, \sigma^2)$$

All of the sample data are from the **same** distribution and the draw of the ith data point is **unrelated** or **independent** of any other data point

Prediction in the Normal Model

- Knowing the values of μ and σ, what is the best predictor of Y_f, a new sample data point?
  - It has to be μ
- How accurate will those predictions be?
  - We can construct a prediction interval that contains the new observation 95% of the time
  - This interval should depend on σ! Why?
  - In this example, a 95% prediction interval is:

$$(\mu - 1.96 \times \sigma, \mu + 1.96 \times \sigma)$$

    - $\mu - 1.96 \times \sigma$ is the 2.5th percentile of $\mathcal{N}(\mu, \sigma^2)$
    - $\mu + 1.96 \times \sigma$ is the 97.5th percentile of $\mathcal{N}(\mu, \sigma^2)$
  - We often write the interval as

$$\mu \pm 1.96\sigma = \mu \pm z_{0.975}\sigma$$

Prediction: The Chip Example

- **Predict one chip:** Before testing a new chip from the same process, what performance score should we predict?
  - Assuming we know $\mu = 100$ and $\sigma = 10$, our point prediction is the population mean, **100 points**
- **Describe uncertainty:** A 99% prediction interval is:

$$\mu \pm 2.58\sigma = 100 \pm 2.58(10) = (74.2,\ 125.8),$$

  where 2.58 is approximately the 99.5th percentile of a standard normal distribution. Under this model, about 99% of chips have scores in this range
- What proportion of chips have performance scores above 80.4?

Chip Binning: The Apple Example

- Manufacturers test chips and sort them by their capabilities, a process called **chip binning**
- Apple's **A12X** and **A12Z** share the same underlying chip design, with different numbers of enabled GPU cores: **7 in the A12X** and **8 in the A12Z**



A12X		A12Z
CPU 4 P-core (2.49 GHz) 4 E-core (1.59 GHz)	+0% +0%	CPU 4 P-core (2.49 GHz) 4 E-core (1.59 GHz)
Neural Engine 8 core (5 TOPS)	+0%	Neural Engine 8 core (5 TOPS)
GPU 7 core (1125 Mhz) 28 EU / 448 ALU (1.008 TFLOPS)	+14%	GPU 8 core (1125 Mhz) 32 EU / 512 ALU (1.152 TFLOPS)
RAM 6 Gb (2133 MHz)	+0%	RAM 6 Gb (2133 MHz)

Prediction and Product Planning

- **Return to our score model:** Recall that $Y \sim \mathcal{N}(100, 10^2)$. Suppose scores of at least **110** qualify for a premium product; the rest go into a standard product
- About **16%** qualify for the premium tier: roughly **1600 of 10,000 working chips**, on average
  - 110 is one standard deviation above the mean of 100, so it is approximately the 84th percentile of $\mathcal{N}(100, 10^2)$
- Predicting these quantities helps the company plan how many products of each type it can supply and how many other components to order

Estimation of the Mean and Standard Deviation

Estimation of the Mean and Standard Deviation

- In practice, we usually don't know μ or σ. We only observe **i.i.d.** samples: $Y_1, Y_2, \ldots, Y_N$, where N is the sample size
  - Example: We do not know the average performance score (μ) or how much scores vary (σ) across chips from a production process. We test a sample of chips and use their scores to estimate these quantities
- Suppose that all we know is that the distribution is normal. We want to use these samples to make inferences about μ and σ

Estimation of the Mean and Standard Deviation

- How do we estimate the parameters of the normal distribution ("population parameters"), μ and σ?
- Sample Mean:

$$\bar{Y} = \frac{1}{N}Y_1 + \frac{1}{N}Y_2 + \cdots + \frac{1}{N}Y_N = \sum_{i=1}^{N} \frac{1}{N}Y_i$$

- Sample Standard Deviation:

$$s_Y = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(Y_i - \bar{Y})^2}$$

Accuracy of the Parameter Estimates

- Our sample estimates of the population parameters can be inaccurate
  - Because of **sampling errors**
- With statistical models, we can estimate the accuracy of these estimates
- From now on, we will focus on the sample mean as an estimator of the normal mean and on the accuracy of that estimate

Accuracy of the Sample Mean: The Role of σ

- How accurate is $\bar{Y}$ as an estimate of μ? How far away can $\bar{Y}$ get from μ, the population parameter $\bar{Y}$ is designed to estimate?
- The sample mean can be written as

$$\bar{Y} = \frac{1}{N}Y_1 + \frac{1}{N}Y_2 + \cdots + \frac{1}{N}Y_N = \sum_{i=1}^{N} \frac{1}{N}Y_i$$

- Subtract μ from both sides of the above equation

$$\bar{Y} - \mu = \frac{1}{N}(Y_1 - \mu) + \frac{1}{N}(Y_2 - \mu) + \cdots + \frac{1}{N}(Y_N - \mu)$$

- How far $Y_i, i = 1 \ldots N$ can deviate from μ depends on σ, which also governs how far $\bar{Y}$ can deviate from μ
- **Takeaway 1**: If the data are "noisy" or highly variable (with large σ), the sample mean may not be a precise estimate of μ

Accuracy of the Sample Mean: The Role of Sample Size N

- **Takeaway 2**: When the sample size N is large, the sample mean is an accurate estimator of the normal mean μ
- Intuition: Each observation Y_i is independent of the others and is distributed symmetrically about μ. Thus, there is a lot of cancellation in this sum. As N gets larger, there is more cancellation and the sample average converges or gets "close" to μ with higher and higher probability

Distribution of $\bar{Y}$

- $\bar{Y}$ is a linear combination of normal random variables and, therefore, is also normally distributed
- The expectation and variance of $\bar{Y}$ are

$$E(\bar{Y}) = \frac{1}{N}\sum_{i=1}^{N} E(Y_i) = \mu$$

$$Var(\bar{Y}) = Var(\sum_{i=1}^{N} \frac{1}{N} Y_i) = \sum_{i=1}^{N} (\frac{1}{N})^2 Var(Y_i) = \frac{1}{N}\sigma^2$$

  - Note that we assume that the data are from a random sample. When $Y_1 \ldots Y_N$ are independent, the variance of the sum is the sum of the variances

- We can conclude that

$$\bar{Y} \sim \mathcal{N}(\mu, \frac{\sigma^2}{N})$$

This means that $\bar{Y}$ is more likely to be a precise estimator of μ with small σ and large N

Law of Large Numbers

- The Law of Large Numbers says that as the sample size N increases, the sample mean will converge to the true mean of the underlying probability model
  - In the normal distribution example: as N increases, $\bar{Y}$ will become a more precise estimator of μ
- This rule applies to distributions other than the normal distribution
  - For any random variable Y that follows a distribution with standard deviation σ_Y, we have

$$Var(\bar{Y}) = Var(\sum_{i=1}^{N} \frac{1}{N} Y_i) = \sum_{i=1}^{N} (\frac{1}{N})^2 Var(Y_i) = \frac{1}{N} \sigma_Y^2$$

    This is the same variance formula as in the normal example

- Key takeaway: **Large samples give more reliable estimates**

Standard Error

- Now, we come back to the example with normal distribution, so

$$\bar{Y} \sim \mathcal{N}(\mu, \frac{\sigma^2}{N})$$

- The standard error of $\bar{Y}$ is its standard deviation across repeated samples of the same size. In other words, different samples *of the same size* give different estimates. Standard error describes how much those estimates vary
- Given the definition, the standard error is

$$\frac{\sigma}{\sqrt{N}}$$

- However, σ is usually unknown. Therefore, we estimate this standard error from the sample

Standard Error: Computation

- How do we estimate the standard error of $\bar{Y}$?
  - Compute the **sample standard deviation** s_Y
  - Divide it by $\sqrt{N}$ to obtain the estimated standard error

$$s_{\bar{Y}} = \frac{s_Y}{\sqrt{N}}$$

- **Difference between s_Y and $s_{\bar{Y}}$:**
  - s_Y describes variation among individual observations
  - $s_{\bar{Y}}$ estimates variation in sample means across repeated samples

Example: Chip Production

- We want to estimate μ, the average performance score of chips produced by a manufacturing process
- We randomly sample five chips and record their performance scores:

$$93,\ 95,\ 100,\ 105,\ 117$$

- **Q1:** What is your estimate of the population mean μ?
  - **Answer:** $\bar{Y} = (93 + 95 + 100 + 105 + 117)/5 = \mathbf{102}$
- **Q2:** What is our estimate of the population standard deviation?
  - **Answer:** $s_Y = \sqrt{\frac{\sum_{i=1}^{5}(Y_i - \bar{Y})^2}{5-1}} = \sqrt{\frac{368}{4}} \approx \mathbf{9.59}$
- **Q3:** What is the standard error of $\bar{Y}$, our estimate of the population mean?
  - **Answer:** $s_{\bar{Y}} = s_Y/\sqrt{N} = 9.59/\sqrt{5} \approx \mathbf{4.29}$

Confidence Intervals and Significance

Confidence Intervals

- We talked about how $\bar{Y}$ can be an accurate estimator of μ, up to some statistical uncertainty governed by σ and N
  - This uncertainty arises from sampling error
- We can construct a confidence interval around the sample mean to reflect this statistical uncertainty at some confidence level
- **Interpretation**: A confidence interval with a 95% confidence level means that 95% of the time, this interval will cover the true value of the unknown parameter μ
- **It's different from a prediction interval:** A confidence interval is about a parameter that governs the population, while a prediction interval is about a feature of one or a few unobserved objects

Deriving a Confidence Interval: Normal Distribution Example

- Recall:

$$\bar{Y} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{N}\right)$$

  and $s_{\bar{Y}}$ estimates the standard deviation of $\bar{Y}$, i.e. $\frac{\sigma}{\sqrt{N}}$

- When the sample size $N > 30$, $s_{\bar{Y}}$ is a reliable approximation to $\frac{\sigma}{\sqrt{N}}$
  - For $N \leq 30$, we will revisit this case when we introduce the t-distribution

Deriving a Confidence Interval: Normal Distribution Example

- Approximating $\frac{\sigma}{\sqrt{N}}$ with $s_{\bar{Y}}$, we find that in repeated sampling, $\bar{Y}$ will lie within the following range 95% of the time:

$$\mu - 1.96 \times s_{\bar{Y}} \leq \bar{Y} \leq \mu + 1.96 \times s_{\bar{Y}}$$

  - Why ± 1.96?
  - We call these cutoff values ***z*-scores**
  - A *z*-score tells us how many standard deviations away from the mean we must go to reach the $100p$th percentile
  - For example: $z_{0.025} = -1.96$ (2.5th percentile), $z_{0.975} = 1.96$ (97.5th percentile)

- Rearranging, this is equivalent to saying that μ will fall within the following interval 95% of the time:

$$\left(\bar{Y} - 1.96 \times s_{\bar{Y}}, \bar{Y} + 1.96 \times s_{\bar{Y}}\right)$$

Confidence Level

- What if we want to be more confident? We can choose a higher confidence level, such as 99%
- 99% Confidence Interval:

$$\left(\bar{Y} - 2.58 \times s_{\bar{Y}}, \bar{Y} + 2.58 \times s_{\bar{Y}}\right)$$

  - $z_{0.005} = -2.58, z_{0.995} = 2.58$
- In R, we can use the function `qnorm()` to obtain the multiplier for $s_{\bar{Y}}$
  - `qnorm(p)` returns the $100p$th percentile of the standard normal distribution $\mathcal{N}(0, 1)$
  - Example: `qnorm(0.025)` $= -1.96$ (2.5th percentile);
    `qnorm(0.005)` $= -2.58$ (0.5th percentile)

Significance Level

- Related to the concept of confidence is the concept of a "significance" level
  - Later in the module, we will review significance in the context of hypothesis testing
- The significance level α is defined as

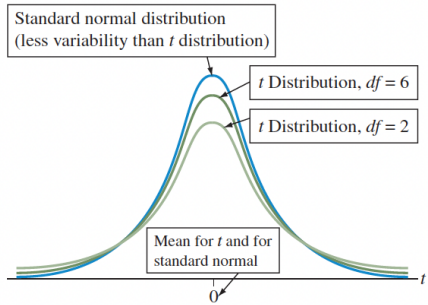
$$\alpha = 1 - \text{Confidence Level}$$

where α is the significance level

- Typical values of α are .05, .01, and .001

Small Samples and the *t*-Distribution

- When the sample size is small ($N \leq 30$), $s_{\bar{Y}}$ may not be a very accurate estimator of $\frac{\sigma}{\sqrt{N}}$
- We use ***t*-scores** instead of *z*-scores to construct confidence intervals
- *t*-scores come from *t*-distributions, which are similar to the standard normal distribution $\mathcal{N}(0,1)$ but with heavier tails
  - How heavy the tails are depends on the number of **degrees of freedom** *df*
  - Large *df* $\implies$ *t*-distribution closer to $\mathcal{N}(0,1)$
  - For samples of size $N \leq 30$, we want to use *t*-scores with $df = N - 1$
  - A *t* critical value is a percentile of the *t*-distribution that we multiply by the estimated standard error to construct a confidence interval



Small Samples and the t-Distribution: Chip Production

- In the five-chip example above, the sample mean is $\bar{Y} = 102$ and the estimated standard error is $s_{\bar{Y}} = 4.29$
- The 95% confidence interval for μ can be constructed as

$$\left(\bar{Y} - 2.776 \times s_{\bar{Y}},\ \bar{Y} + 2.776 \times s_{\bar{Y}}\right)$$

  - With $df = N - 1 = 4$, $t^4_{0.025} \approx -2.776$ and $t^4_{0.975} \approx 2.776$
  - It is wider than the corresponding confidence interval for a large sample size N
  - Substituting the values gives the 95% confidence interval $(90.09, 113.91)$
  - What can we do to get a tighter confidence interval?
- In R, we can use the function `qt()` to obtain the required t-scores:
  - `qt(p, df)` returns the $100p$th percentile of the t-distribution with `df` degrees of freedom
  - Example: `qt(0.975, df = 4)` ≈ 2.776

Example: Chip Production and Prediction Interval

- We have estimated the population mean. Now, what performance score should we predict for the next chip?
- Our **point prediction** is the sample mean: $\bar{Y} = 102$
- Assuming the next chip is independent of our sample and comes from the same normal population, a 95% prediction interval is

$$\bar{Y} \pm t_{0.975,N-1}\, s_Y \sqrt{1 + \frac{1}{N}} \approx (72.83,\ 131.17)$$

  - This interval is wider than the confidence interval for μ: we must account for both uncertainty about the population mean and variation among individual chips
  - A confidence interval accounts only for uncertainty about the population mean
- You do not need to memorize this formula; focus on the difference between estimating the population mean and predicting one chip's performance

Central Limit Theorem (CLT)

CLT: Key Idea and Why It Matters

- In practice, many data are not generated following a normal distribution (e.g., income)
  - Can we still construct confidence intervals for population means using sample statistics
- The Central Limit Theorem (CLT) says:

  Even if you're not normal, your sample average is (approaches) normal
- This powerful theorem allows us to apply inference methods as in the normal mean distribution example

CLT Illustration: Uniform Distribution

- Draw $N = 10$ independent observations from $U[0, 1]$ and compute their mean
- Repeat this experiment 10, 100, and 10,000 times
- Compare the histograms; each sample still has $N = 10$ observations

```r
set.seed(123)  # reproducibility

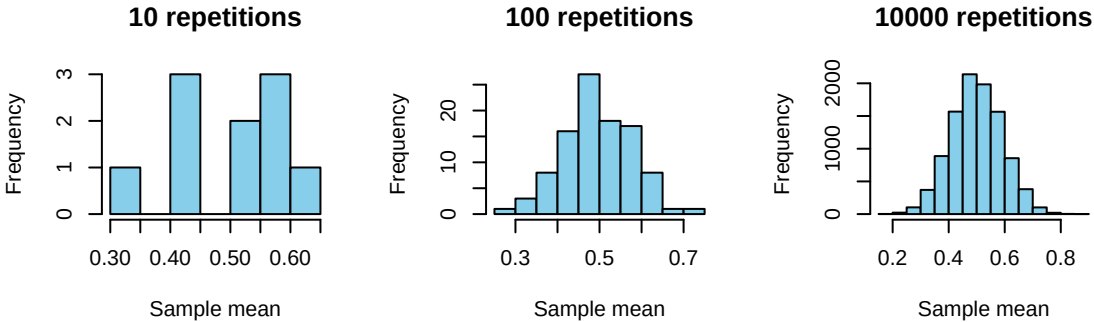
# Function to generate sample means
get_means <- function(n_rep, n_draws=10) {
  replicate(n_rep, mean(runif(n_draws, 0, 1)))
}

means_10   <- get_means(10)      # 10 repetitions
means_100  <- get_means(100)     # 100 repetitions
means_10000 <- get_means(10000) # 10000 repetitions

par(mfrow=c(1,3))  # arrange plots in 1 row, 3 columns
hist(means_10, breaks=5, col="skyblue",main="10 repetitions", xlab="Sample mean")
hist(means_100, breaks=10, col="skyblue",main="100 repetitions", xlab="Sample mean")
hist(means_10000, breaks=20, col="skyblue",main="10000 repetitions", xlab="Sample mean")
```

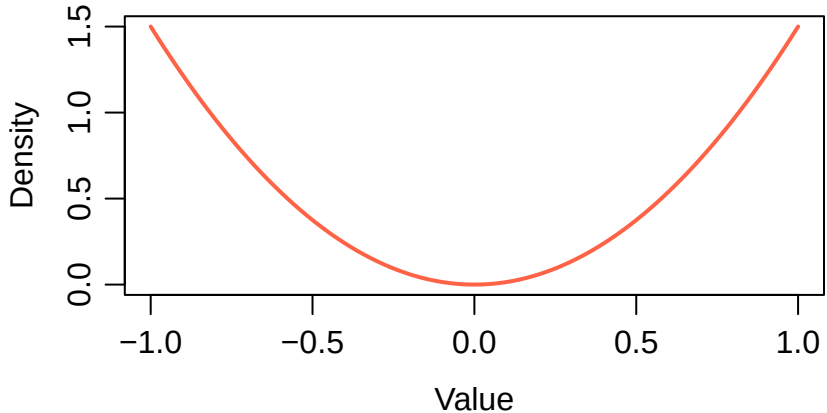
CLT Illustration: Uniform Distribution

- More repetitions give a clearer histogram of the **same sampling distribution**. Increasing the sample size *N* improves the normal approximation



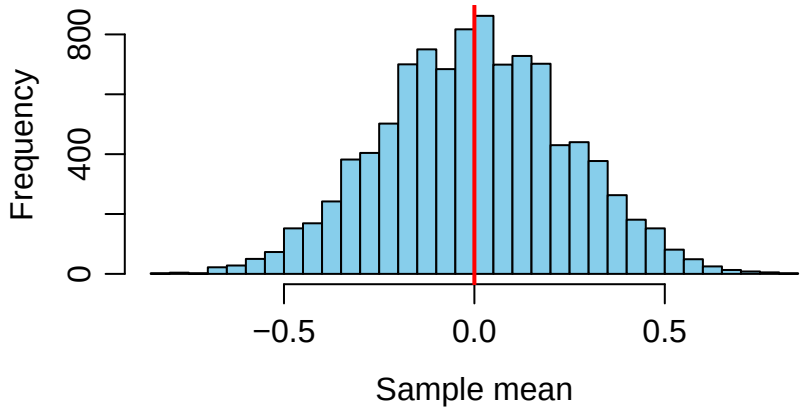
CLT Illustration: A Heavy-Tailed Distribution

- Now, let's see if the same will happen with a more unusual distribution with U-shapes, i.e., very spread out



CLT Illustration: A Heavy-Tailed Distribution

- The distribution of sample means still looks normal!



CLT: Intuition and Takeaway

- **Intuition**: Each sample mean involves the summation of independent random variables. Random fluctuations from the mean tend to **cancel out**, leaving a "normal" distribution centered around the true mean
- **Caveat**: The sample mean “approaches” a normal distribution with a growing sample size. With small N, the distribution of the sample mean may deviate substantially from normal
  - For $N \leq 30$, we should consider t-distribution instead of normal distribution, as we did for constructing confidence intervals for $\hat{Y}$ from normal distribution
  - For small samples from a strongly nonnormal population, a t-interval may still be unreliable
- **Takeaway**: Even if the underlying probability model does not follow normal distribution, with reasonably large sample size, we can still investigate statistical uncertainty and construct confidence interval for population mean using sample statistics, as we did in the normal distribution example

Hypothesis Testing

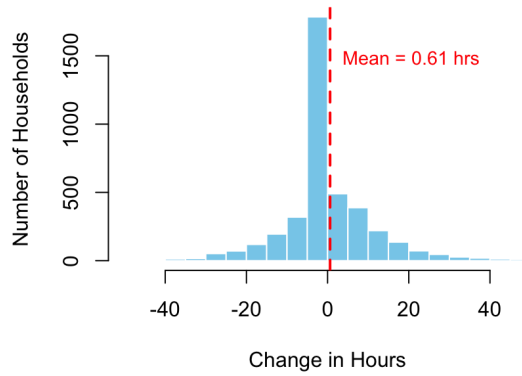
Netflix: The Business Question

- Suppose Netflix pilots a beta app with built-in games to encourage more usage
- We observe weekly Netflix usage for 4,000 households before and after the launch, including households without a Netflix subscription
- A manager claims that the average increase across all U.S. households is **0.5 hour per week**
- **Question:** Do the data provide evidence that the population mean increase differs from 0.5 hour?



Netflix: Understanding the Data

- For this exercise, assume the households form an independent, representative sample of all U.S. households
- Each household contributes **one change**: after-period usage minus before-period usage
- The histogram of household changes is **not normal**
- With $N = 4000$, the CLT supports an approximately normal sampling distribution of the **sample mean**, assuming finite variance



Netflix: Summarizing the Sample

```r
streaming_beta <- read.csv("../../Data/streaming_beta.csv")
hours_change <- streaming_beta$hours_post - streaming_beta$hours_pre

N <- length(hours_change)
mean_change <- mean(hours_change)
se <- sd(hours_change) / sqrt(N)

cat("Number of households:", N, "\n")
## Number of households: 4000
cat("Mean change:", round(mean_change, 3), "hours\n")
## Mean change: 0.607 hours
cat("Estimated standard error:", round(se, 3), "hours\n")
## Estimated standard error: 0.174 hours
```

- The sample mean is about **36.4 minutes**, compared with the manager's claim of 0.5 hour
- Is this difference large relative to the sampling uncertainty?

Step 1: State the Hypotheses

- **Hypothesis testing** evaluates a claim about the **population mean**, using the **sample mean** as evidence
- Let Y_i be household i's change in weekly Netflix usage: after minus before, measured in hours
- Let μ be the population mean of these changes across all U.S. households
- The **null hypothesis** (H_0) is the benchmark claim we test against the **alternative hypothesis** (H_1)

$$H_0 : \mu = 0.5 \qquad H_1 : \mu \neq 0.5$$

- This is a **two-tailed test**: an increase either below or above 0.5 hour would differ from the manager's claim
- **A hypothesis test cannot prove that H_0 is true**: failing to reject H_0 only means that we do not have enough evidence against it

Step 2: Measure the Distance from the Null

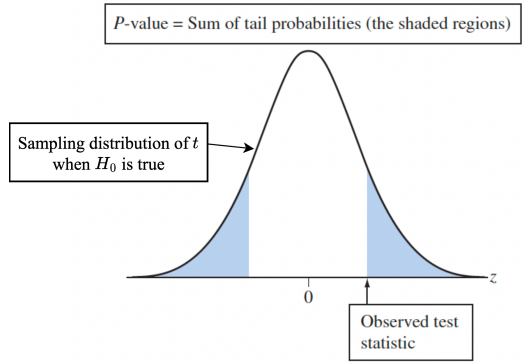
- The sample mean will usually differ from the null value because of sampling error, even when H_0 is true
- We measure this difference relative to the **estimated standard error** of the sample mean

$$t = \frac{\bar{Y} - \mu_0}{s_Y / \sqrt{N}}$$

- Here, $\mu_0 = 0.5$ is the mean specified by the null hypothesis, and s_Y is the sample standard deviation of household changes
- The ***t*-statistic** tells us how many estimated standard errors the sample mean is above or below the null value
- For a two-tailed test, a larger $|t|$ provides stronger evidence against H_0

Step 2: The p-Value

- **Assuming H_0 is true**, how likely is a test statistic at least as extreme as the one we observed?
- This probability is the **p-value**
- For a two-tailed test, count outcomes at least as far from zero in **either direction**
- A small p-value means that such an extreme result would be unusual under H_0



Step 3: Make a Decision

- Choose a **significance level** before examining the results, such as $\alpha = 0.05$
- If the p-value is **below** α, reject H_0 in favor of H_1
- If the p-value is **at least** α, there is not enough evidence to reject H_0
- A 5% significance level means that, when H_0 is true and the test's assumptions hold, the procedure rejects H_0 about 5% of the time across repeated samples
- The p-value does **not** tell us the probability that H_0 is true

Netflix: Calculating the Two-Tailed p-Value

- Use a one-sample t-test with $N - 1$ degrees of freedom
- Since household changes are not normal, this test is **approximate** here, supported by the large sample size

```
t_stat <- (mean_change - 0.5) / se
p_value <- 2 * pt(-abs(t_stat), df = N - 1)

cat("t-statistic:", round(t_stat, 3), "\n")
## t-statistic: 0.617
cat("Two-tailed p-value:", round(p_value, 3), "\n")
## Two-tailed p-value: 0.537
```

- `abs()` takes the absolute value, so `pt(-abs(t_stat), df = N - 1)` gives the lower-tail probability
- By symmetry, the upper tail has the same probability, so we multiply by 2 to count **both tails**

Netflix: The Two-Tailed Test in R

- `t.test()` calculates the test statistic, p-value, and confidence interval

```
two_tailed <- t.test(hours_change, mu = 0.5,
                     alternative = "two.sided", conf.level = 0.95)
two_tailed
##
##  One Sample t-test
##
## data:  hours_change
## t = 0.61747, df = 3999, p-value = 0.537
## alternative hypothesis: true mean is not equal to 0.5
## 95 percent confidence interval:
##  0.2664459 0.9483041
## sample estimates:
## mean of x
##  0.607375
```

Netflix: The Two-Tailed Conclusion

- The p-value is 0.537, which is greater than 0.05: we **do not reject** $H_0 : \mu = 0.5$
- There is not enough evidence that the population mean increase differs from 0.5 hour per household per week
- The corresponding 95% confidence interval is [0.266, 0.948] hours
  - The null value of 0.5 hour lies inside this interval, consistent with the test result
- Failing to reject H_0 **does not prove** that the population mean increase is exactly 0.5 hour

Netflix: A Different Business Question

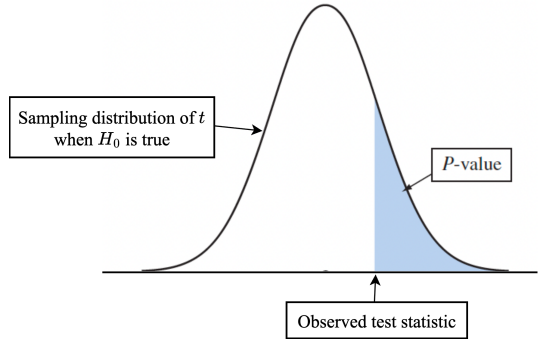
- Now suppose Netflix requires an increase of more than **0.25 hour per household per week** to justify the investment
  - In practice, we often use "hurdle rate" to decide whether a business strategy will be taken
- **Question:** Is there evidence that the population mean increase **exceeds 0.25 hour**?
- What are H_0 and H_1?

$$H_0 : \mu \leq 0.25 \qquad H_1 : \mu > 0.25$$

- This is a **one-tailed test**: evidence in the positive direction supports H_1

One-Tailed Test: The p-Value

- Calculate the test statistic and p-value at the **boundary of the null hypothesis**: $\mu = 0.25$
- For $H_1 : \mu > 0.25$, the p-value is the probability of a test statistic **at least as large** as the one observed
- Only the **upper tail** counts in this test
- Choose the threshold and direction based on the business question **before examining the data**



Netflix: The One-Tailed Test in R

- Use `alternative = "greater"` to test if the population mean exceeds 0.25 hours

```
one_tailed <- t.test(hours_change, mu = 0.25,
                     alternative = "greater", conf.level = 0.95)
one_tailed
##
##  One Sample t-test
##
## data:  hours_change
## t = 2.0551, df = 3999, p-value = 0.01997
## alternative hypothesis: true mean is greater than 0.25
## 95 percent confidence interval:
##  0.3212786       Inf
## sample estimates:
## mean of x
##  0.607375
```

- The p-value is about 0.020, which is below 0.05: we **reject** $H_0 : \mu \leq 0.25$
- There is evidence that the population mean increase **exceeds 0.25 hour** per household per week

Extension: Comparing Two Groups

- A **one-sample test** compares a population mean with a specified value
- A **two-sample test** compares the means of two populations using data from two groups
  - Example: comparing average spending between two independently sampled customer groups
  - We will not cover the details of two-sample tests, but you can also run them using `t.test()`
- Next, we apply hypothesis testing to mutual fund returns

Stock Returns Data Example

The Mutual Fund Data

- The `Vanguard` dataset in the `DataAnalytics` package contains monthly returns on various Vanguard mutual funds

$$R_t = \frac{(P_t - P_{t-1}) + D_t}{P_{t-1}}$$

If you purchase one share at time $t-1$ and liquidate your position at time t, you earn capital gains plus any dividends

The Mutual Fund Data

```
data(Vanguard, package="DataAnalytics")
head(Vanguard)
##         date ticker crsp_fundno   mtna      mret
## 1 1988-04-29  VEIPX       31217     NA  0.010215
## 2 1988-05-31  VEIPX       31217     NA  0.027300
## 3 1988-06-30  VEIPX       31217 16.763  0.030535
## 4 1988-07-29  VEIPX       31217     NA  0.005797
## 5 1988-08-31  VEIPX       31217     NA -0.013449
## 6 1988-09-30  VEIPX       31217 28.259  0.041992
```

- There are 9 mutual funds in the dataset
- The variable `ticker` tells us which fund the data is on
- The variable `mret` contains the monthly returns for each fund

The Mutual Fund Data

- Returns for all 9 mutual funds are stacked on top of each other in one column
- Let's "unstack" or `reshape` the data so that there is an array with one column for each fund. That is, let's create a new dataset with 10 columns (the date and monthly returns for each of the 9 funds)
- To do this, we will use the R contributed package `reshape2` available from CRAN, specifically, the `dcast` function to take the "molten" data and cast in in a specific shape (mold)

The Mutual Fund Data

```
install.packages("reshape2") #do this once, if necessary
```

```
library(reshape2) #load the package
Van = Vanguard[ , c(1,2,5)] #grab relevant cols
V_reshaped = dcast(Van, date~ticker, value.var="mret") #reshaped dataset
```

```
head(V_reshaped, n=5)
##         date VEIPX VFIAX     VGENX     VGHCX VMGRX VQNPX VSMAX VTSAX VWNFX
## 1 1984-06-29    NA    NA -0.067921  0.002070    NA    NA    NA    NA    NA
## 2 1984-07-31    NA    NA -0.105381 -0.021694    NA    NA    NA    NA    NA
## 3 1984-08-31    NA    NA  0.186717  0.090813    NA    NA    NA    NA    NA
## 4 1984-09-28    NA    NA  0.016895 -0.023233    NA    NA    NA    NA    NA
## 5 1984-10-31    NA    NA -0.041537  0.027750    NA    NA    NA    NA    NA
```

```
dim(Vanguard)
## [1] 2294    5
dim(V_reshaped)
## [1] 349  10
```

The Mutual Fund Data: Summary Statistics

- Let's compute some simple summary statistics

```
library(DataAnalytics)
descStat(V_reshaped)
##         Mean Median    SD   IQR SE Mean 95% CI-L 95% CI-U NMissing
## VEIPX 0.009  0.012 0.037 0.043   0.002    0.005    0.013       46
## VFIAX 0.004  0.011 0.045 0.050   0.004   -0.003    0.011      198
## VGENX 0.012  0.012 0.060 0.073   0.003    0.005    0.018        0
## VGHCX 0.014  0.016 0.041 0.046   0.002    0.010    0.018        0
## VMGRX 0.010  0.013 0.072 0.073   0.005    0.000    0.021      163
## VQNPX 0.009  0.014 0.045 0.055   0.003    0.004    0.014       31
## VSMAX 0.009  0.017 0.059 0.072   0.005    0.000    0.018      198
## VTSAX 0.005  0.012 0.046 0.053   0.004   -0.003    0.012      198
## VWNFX 0.009  0.014 0.043 0.048   0.002    0.005    0.014       13
## Number of Observations =  349
```

- Why are there missing observations?

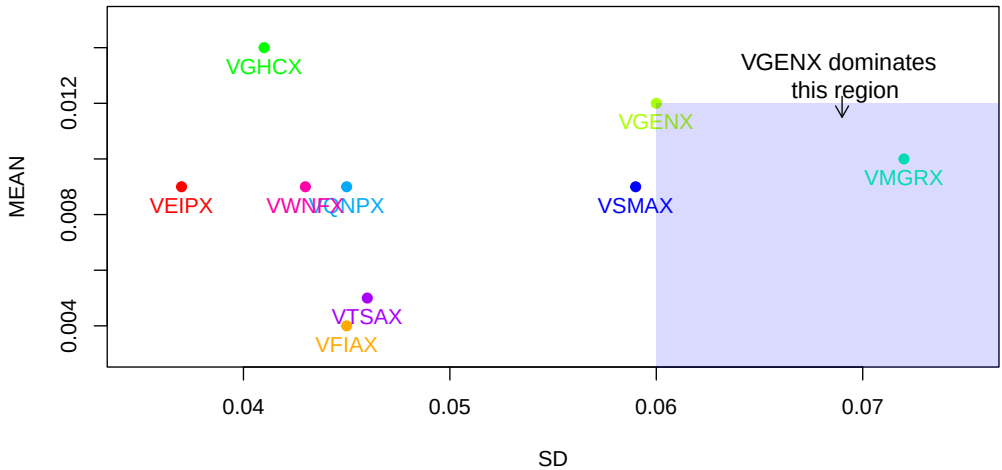
The Mutual Fund Data

- Let's plot standard deviation vs. the mean for each fund

```
mat = descStat(V_reshaped) # save the descriptive stats
SD   = mat[ , 3]           # fetch relevant col for sd
MEAN = mat[ , 1]           # fetch relevant col for mean
FUNDS = rownames(mat)      # retrieve fund ticker symbols

plot(x=SD, y=MEAN, pch=19, col=rainbow(9))
text(x=SD, y=MEAN, labels=FUNDS, pos=1, col=rainbow(9))
```

The Mutual Fund Data: Plot



The Mutual Fund Data: Variability

```
##        Mean Median    SD   IQR SE Mean 95% CI-L 95% CI-U NMissing
## VEIPX 0.009  0.012 0.037 0.043   0.002   0.005   0.013       46
## VFIAX 0.004  0.011 0.045 0.050   0.004  -0.003   0.011      198
## VGENX 0.012  0.012 0.060 0.073   0.003   0.005   0.018        0
## VGHCX 0.014  0.016 0.041 0.046   0.002   0.010   0.018        0
...
```

- What is the difference between `SD` and `SE Mean`?
- `SD`: the variability of the data; `SE Mean`: the variability of mean return estimates

Variability of the Data

Population standard deviation:

$$\text{SDev}(Y) = \sigma_Y$$

Sample standard deviation:

$$s_Y = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(Y_i - \bar{Y})^2}$$

Variability of the Sample Mean

Population standard deviation of $\bar{Y}$:

$$\text{SDev}(\bar{Y}) = \sigma_{\bar{Y}} = \frac{\sigma_Y}{\sqrt{N}}$$

Standard error of the mean:

$$s_{\bar{Y}} = \frac{s_Y}{\sqrt{N}}$$

The Mutual Fund Data: Sampling Error

- Is VEIPX better than VFIAX?
  - Mean is much higher
  - SD is about the same
  - But look at the confidence intervals – they overlap!

```
##        Mean Median    SD   IQR SE Mean 95% CI-L 95% CI-U NMissing
## VEIPX 0.009  0.012 0.037 0.043   0.002    0.005    0.013       46
## VFIAX 0.004  0.011 0.045 0.050   0.004   -0.003    0.011      198
## Number of Observations =  349
```

The Mutual Fund Data: Two-sample *t*-Test

- We can use two-sample *t*-test to test if VEIPX is better than VFIAX or not
- Should we use one-tailed or two-tailed test?
- **Conclusion:** We cannot reject that VEIPX performs the same as VFIAX

```
##
##  Welch Two Sample t-test
##
## data:  V_reshaped$VEIPX and V_reshaped$VFIAX
## t = 1.1304, df = 252, p-value = 0.1297
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  -0.002204049          Inf
## sample estimates:
##   mean of x   mean of y
## 0.008746119 0.003959993
```

Appendix

Glossary of R Commands

- `c()` Combine values into a vector or list
- `sd(sample)` returns sample standarde deviation
- `data(A)` allows access to data frame A that is in the data library
- `head(A)` Returns the first parts of a data frame A
- `t.test(sample, alternative = c("two.sided", "less", "greater"), mu)` Conducts t test
- `library("PERregress")` loads the R package containing datasets and customized functions for our class
- `str(object)` tells you the "structure" of an object, e.g. str(dataframe)
- `descStat(A)` produce descriptive statistics for all the variables in data frame A that are not factors
- `qnorm(p)` returns the 100pth percentile of the standard normal distribution $\mathcal{N}(0, 1)$
- `qt(p, df)` returns the 100pth percentile of the t-distribution with `df` degrees of freedom